Lizhi Yang

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Research Interests

My research aims to create robust, safe, and intelligent robots systems that can operate reliably in the real world. I focus on the intersection of loco-manipulation, robot safety, whole-body control, and perception-enabled control.

Education

California Institute of Technology — **Pasadena, CA**
Ph.D. Candidate, Mechanical Engineering (Robotics, Controls and Dynamics) — *Sept 2022 – Present*

California Institute of Technology — **Pasadena, CA**
M.S. Mechanical Engineering (along the way) — *June 2024*

University of California, Berkeley — **Berkeley, CA**
B.S. Electrical Engineering and Computer Sciences — *May 2022*

Professional Experience

California Institute of Technology — **Pasadena, CA**
Ph.D. Candidate — *2022 – Current*

- Developed MorphMimic, sharing one morphology-conditioned world model and motion-tracking policy across Unitree G1, Agibot X2 Ultra, and Booster T2, achieving over 99.7% clip completion in simulation and deploying on the Unitree G1 for motion tracking and teleoperation.
- Developed a training-free contextual safety filter for text-conditioned motion generators, grounding natural-language safety rules in reference trajectories enforced via a CBF-QP, validated on the Unitree G1.
- Developed mobile sensor deployment system based on quadrupedal mobile manipulator, enabling autonomous sensor deployment in the wild.
- Built full-stack pipeline of humanoid robot whole-body task-space controller, enabling the Unitree G1 to complete natural language and vision driven tasks in the real world with no task data collection.
- Developed a perception-aware CBF-RL framework integrating real-time vision and control barrier functions to enable safe whole-body safety.
- Constructed social-semantic safety frameworks for general mobile robot platforms, validated on the Unitree G1 and Go2.
- Developed pipelines for data collection, finetuning, and deployment of VLA models for Franka Emika.
- Designed reinforcement learning and hardware deployment pipeline for humanoid robots, enabling real-world experiments.
- Developed methods to enforce safety in learned control and planning policies, improving robustness of autonomous systems.
- Created push-recovery and uncertainty-aware locomotion frameworks, validated on humanoid platforms.

University of California, Berkeley — **Berkeley, CA**
Student Research Assistant — *2019 – 2022*

- Conducted robotics research on locomotion, perception, and multi-robot collaboration, with deployments on Unitree A1 and MIT Mini Cheetah.
- Built multi-robot system frameworks integrating LiDAR and depth sensing for collaborative load-moving tasks.
- Developed vision-based navigation and obstacle detection stacks in ROS, enabling autonomous quadrupedal navigation.
- Applied reinforcement learning (PPO) and Bayesian optimization for locomotion control, improving stability and adaptability.

Samsara Inc. — **San Francisco, CA**
Software Engineering Intern — *2021*

- Delivered production features in a 4-person engineering team, contributing to fleet management products.
- Built machine learning benchmarking tools to compare models at scale, improving workflow efficiency for ML engineers.
- Leveraged C++ and Go for backend development and performance optimization.

Awards & Honors

2025: Best Oral Presentation Finalist, International Conference on Humanoid Robots (Humanoids)

2022: Arthur M. Hopkin Award – University of California, Berkeley

2021: Best Service Robotics Paper Finalist, International Conference on Robotics and Automation (ICRA)

Publications (* denotes equal contribution)

[1] ***Lizhi Yang**, *Blake Werner, Massimiliano de Sa, and Aaron D. Ames. "CBF-RL: Safety Filtering Reinforcement Learning in Training with Control Barrier Functions". In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2026.

[2] Min Dai, William D. Compton, Junheng Li, **Lizhi Yang**, and Aaron D. Ames. "Walk the PLANC: Physics-Guided RL for Agile Humanoid Locomotion on Constrained Footholds". In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2026.

[3] Junheng Li, **Lizhi Yang**, and Aaron D. Ames. "MIRROR: Visual Motion Imitation via Real-time Retargeting and Teleoperation with Parallel Differential Inverse Kinematics". In: *arXiv:2603.23995*. Preprint. 2026.

[4] Junheng Li, Liang Wu, Sergio A. Esteban, **Lizhi Yang**, Ján Drgoňa, and Aaron D. Ames. "Accelerating and Scaling MPC-Guided Reinforcement Learning for Humanoid Locomotion and Manipulation". In: *arXiv:2606.05687*. Preprint. 2026.

[5] **Lizhi Yang**, Ryan M. Bena, Meg Wilkinson, Gilbert Bahati, Andy Navarro Brenes, Ryan K. Cosner, and Aaron D. Ames. "Safe-SAGE: Social-Semantic Adaptive Guidance for Safe Engagement through Laplace-Modulated Poisson Safety Functions". In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2026.

[6] **Lizhi Yang**, Yiling Hou, Yao Tang, Junheng Li, Daniel Weng, Blake Werner, and Aaron D. Ames. "CSF: Contextual Safety Filtering for Motion Generators". In: *arXiv:2610.12467*. Preprint. 2026.

[7] **Lizhi Yang**, Junheng Li, and Aaron D. Ames. "PAC-MAN: Perception-Aware CBF-RL for Whole-Body Safety in Humanoid Dodgeball". In: *arXiv:2607.28623*. Preprint. 2026.

[8] **Lizhi Yang**, Junheng Li, Nehar Poddar, Yiling Hou, Gio Huh, Robert Griffin, Georgia Gkioxari, and Aaron D. Ames. "HANDOFF: Humanoid Agentic Task-Space Whole-Body Control via Distilled Complementary Teachers". In: *arXiv:2606.06493*. Preprint. 2026.

[9] **Lizhi Yang**, Fengze Xie, Yuyang Wang, Lu Gan, Yisong Yue, and Aaron D. Ames. *MorphMimic: Morphology-Conditioned World Model and Policy Sharing Across Humanoids*. Technical report. 2026. URL: `https://lzyang2000.github.io/morphmimic/`.

[10] Pol Mestres, Shima Sadat Mousavi, Pio Ong, **Lizhi Yang**, Ersin Das, Joel W. Burdick, and Aaron D. Ames. "Explicit Control Barrier Function-based Safety Filters and their Resource-Aware Computation". In: *IFAC World Congress*. 2026.

[11] ***Lizhi Yang**, *Blake Werner, Ryan K. Cosner, David Fridovich-Keil, Preston Culbertson, and Aaron D. Ames. "SHIELD: Safety on Humanoids via CBFs In Expectation on Learned Dynamics". In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2025.

[12] Ryan M. Bena, Gilbert Bahati, Blake Werner, Ryan K. Cosner, **Lizhi Yang**, and Aaron D. Ames. "Geometry-Aware Predictive Safety Filters on Humanoids: From Poisson Safety Functions to CBF-Constrained MPC". In: *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*. 2025.

[13] **Lizhi Yang**, Blake Werner, Adrian B. Ghansah, and Aaron D. Ames. "Bracing for Impact: Robust Humanoid Push Recovery and Locomotion with Reduced Order Models". In: *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*. 2025.

[14] *Blake Werner, ***Lizhi Yang**, and Aaron D. Ames. "Architecture Is All You Need: Diversity-Enabled Sweet Spots for Robust Humanoid Locomotion". In: *arXiv:2510.14947*. Preprint. 2025.

[15] Gilbert Feng, Hongbo Zhang, Zhongyu Li, Xue Bin Peng, Bhuvan Basireddy, Linzhu Yue, Zhitao Song, **Lizhi Yang**, Yunhui Liu, Koushil Sreenath, and Sergey Levine. "Genloco: Generalized Locomotion Controllers for Quadrupedal Robots". In: *Conference on Robot Learning (CoRL)*. PMLR, 2023, pp. 1893–1903.

[16] Xiaoyu Huang, Zhongyu Li, Yanzhen Xiang, Yiming Ni, Yufeng Chi, Yunhao Li, **Lizhi Yang**, Xue Bin Peng, and Koushil Sreenath. "Creating a Dynamic Quadrupedal Robotic Goalkeeper with Reinforcement Learning". In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2023.

[17] Chong Zhang and **Lizhi Yang**. "Generating a Terrain-Robustness Benchmark for Legged Locomotion: A Prototype via Terrain Authoring and Active Learning". In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2023.

[18] ***Lizhi Yang**, *Zhongyu Li, Jun Zeng, and Koushil Sreenath. "Bayesian Optimization Meets Hybrid Zero Dynamics: Safe Parameter Learning for Bipedal Locomotion Control". In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2022.

[19] **Lizhi Yang**, Ruhang Ma, and Avideh Zakhor. "Drone Object Detection Using RGB/IR Fusion". In: *Electronic Imaging: Computational Imaging*. 2022.

[20] Chenyu Yang, Guo Ning Sue, Zhongyu Li, **Lizhi Yang**, Haotian Shen, Yufeng Chi, Akshara Rai, Jun Zeng, and Koushil Sreenath. "Collaborative Navigation and Manipulation of a Cable-Towed Load by Multiple Quadrupedal Robots". In: *IEEE Robotics and Automation Letters* 7.4 (2022), pp. 10041–10048.

[21] Zixian Zang, Haotian Shen, **Lizhi Yang**, and Avideh Zakhor. "Sensor-Aware SLAM-based Frontier Exploration and Mapping". In: *Electronic Imaging: AVM*. 2022.

[22] *Scott Gilroy, *Derek Lau, ***Lizhi Yang**, Ed Izaguirre, Kristen Biermayer, Anxing Xiao, Mengti Sun, Ayush Agrawal, Jun Zeng, Zhongyu Li, and Koushil Sreenath. "Autonomous Navigation for Quadrupedal Robots with Optimized Jumping Through Constrained Obstacles". In: *IEEE CASE*. 2021.

[23] *Anxing Xiao, *Wenzhe Tong, ***Lizhi Yang**, Jun Zeng, Zhongyu Li, and Koushil Sreenath. "Robotic Guide Dog: Leading a Human with Leash-Guided Hybrid Physical Interaction". In: *IEEE International Conference on Robotics and Automation (ICRA)*. Best Paper Award Finalist for Service Robotics. 2021.

[24] **Lizhi Yang**, Ilian Herzi, Avideh Zakhor, Anup Hiremath, Sahm Bazargan, and Robert Tames-Gadam. "Indoor Query System for the Visually Impaired". In: *International Conference on Computers Helping People with Special Needs (ICCHP)*. 2020, p. 517.

[25] Huijuan Xu, **Lizhi Yang**, Stan Sclaroff, Kate Saenko, and Trevor Darrell. "Spatio-Temporal Action Detection with Multi-Object Interaction". In: *EPIC Workshop @ European Conference on Computer Vision (ECCV)*. 2020.

Invited Talks

2025,2026: *From Performance to Safety: Bringing Robots Into the Real World*, Seminar of Prof. Zackory Erickson, Carnegie Mellon University; Seminar of Prof. Xiaobin Xiong, University of Wisconsin–Madison; Seminar of Prof. Adam Wierman, California Institute of Technology

Academic Service

Reviewer: IEEE Transactions on Industrial Electronics (TIE), IEEE Transactions on Robot Learning (TRL) IEEE Robotics and Automation Letters (RA-L), IEEE International Conference on Robotics and Automation (ICRA), IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) and Frontiers in Robotics and AI.

Outreach & Community Service

Pioneers in Engineering PiSens — **UC Berkeley**
Project Manager — *May 2019 – May 2020*
Led team to develop sensor kits for low-cost robot competitions; organized annual competitions for underrepresented high school students in the Bay Area.